A counterexample to the printed perfect-graph substitution speed formula

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Conjecture 00000002195

Precise scope. We use the usual *labelled* speed and ordinary unrestricted single-vertex graph substitution. Both input properties are the *full* hereditary property of finite perfect graphs. The printed statement specifies neither narrower input classes nor a labelled/unlabelled convention. Under this standard universal reading its asserted formula

$$s_n = \exp\big((\tfrac{1}{2} + o(1))n \log n\big)$$

is false. This does not assert a result about an unspecified restricted factorial subclass, an unlabelled speed, or the undefined generating-function clause. The complete original English and Chinese source is included separately.

The actual objects

A finite graph is perfect when every induced subgraph has chromatic number equal to clique number, including the empty graph. In Lean, both numbers are native `SimpleGraph.chromaticNumber` and `SimpleGraph.cliqueNum`. `Perfect` quantifies over all embeddings of every finite labelled vertex set; `perfect_all_induced` proves the corresponding equality for every actual induced vertex subset. The full property is proved hereditary and invariant under genuine graph isomorphisms.

For a graph G, a vertex v, and a graph H, form $G(v \leftarrow H)$ on $(V(G) \setminus \{v\}) \sqcup V(H)$. Adjacency within each part is its original adjacency; an inserted vertex is adjacent to a surviving base vertex x exactly when x was adjacent to v. This is the ordinary operation in Klavžar–Petkovšek, printed p. 333, Example 4(iv). The Lean definition constructs all four adjacency cases and proves symmetry and looplessness. A true graph isomorphism identifies $K_1(v \leftarrow H)$ with H.

Let $\mathcal{P}$ be the full perfect-graph property and let $\mathcal{S}$ consist of all graphs isomorphic to some $G(v \leftarrow H)$ with $G, H \in \mathcal{P}$. The actual product predicate quantifies over all finite sizes, both graphs, the chosen vertex and a genuine graph isomorphism; it does not merely declare a desired family to belong. It is invariant under relabelling. Its labelled speed is

$$s_n = \#\{K \text{ on } \{0, \dots, n-1\} : K \in \mathcal{S}\}.$$

Lean uses the true finite subtype cardinality. Finiteness and $s_n > 0$ are proved, so there is no infinite-cardinality default or unproved logarithm convention.

A true counting lower bound

For every Boolean $m \times m$ matrix A, use the vertex set $\{0, \dots, m-1\} \sqcup \{0, \dots, m-1\}$. Put an edge between the left i and right j precisely when A_{ij} is true, and put no edges within a half. This is an actual simple graph with an actual two-colouring. Every induced graph is two-colourable. Its native chromatic and clique numbers agree: both are 0 in the empty case, 1 in the nonempty edgeless case,

and 2 if an edge exists. Thus each graph is genuinely perfect and belongs to $\mathcal{S}$ through the proved K_1 substitution isomorphism.

Transport along the genuine equivalence of the two halves with $\mathrm{Fin}(m+m)$ gives labelled graphs on $2m$ vertices. Their cross adjacencies recover every entry of A, proving an injection into the actual product subtype. Consequently

$$2^{m^2} \le s_{2m}, \qquad m^2 \log 2 \le \log s_{2m}. \tag{1}$$

These are counts of distinct labelled graphs, not asserted counts of isomorphism classes.

The asymptotic contradiction

Theorem 1. *For every finite real C, there is no eventual bound $\log s_n \le Cn \log n$. In particular, the printed formula is false for this actual substitution product.*

Proof. An eventual bound would hold along the cofinal sequence $n = 2m$. For $m \ge 1$, combine it with (1) and divide by the positive m to obtain

$$\log 2 \le 4C \, \frac{\log(2m)}{2m}.$$

The right side tends to zero, using the native real logarithmic limit transported along the actual natural-number embeddings. Closedness of the order relation then gives $\log 2 \le 0$, a contradiction.

Precisely, the printed formula would provide a real sequence $\varepsilon_n \to 0$ and an eventual equality $s_n = \exp((1/2 + \varepsilon_n) n \log n)$. Eventually $\varepsilon_n < 1/2$ and $n \ge 1$. Taking the actual logarithm gives $\log s_n \le n \log n$, the excluded bound with $C = 1$. The Lean theorem proves this full error-sequence and eventual-equality implication. □

Formalization and reproduction

The complete 275-line proof has 33 explicit declarations: 22 theorems, 10 definitions and one actual instance. All graph, isomorphism, induced-number, finite-cardinality, cast, positivity and filter-limit bridges used above are proved in Lean. The literature supplies standard terminology only; it supplies no Lean axiom. The proof uses no `sorry`, `native_decide` or extra axiom. The namespace audit enumerates all owned declarations, including generated names, and calls ordinary `#print axioms` for each.

The project pins Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d`. From `lean/`, run `lean --version`, then `bash verify.sh` for the build, warning-as-error exhaustive axiom audit and official fresh-kernel replay. The exact commands, original logs, source hashes and execution receipts are included. Production timings are diagnostic, not controlled efficiency evidence.

References

[1] S. Klavžar and M. Petkovšek, *On Characterizations with Forbidden Subgraphs*, Colloquia Mathematica Societatis János Bolyai 52, Combinatorics, Eger (Hungary), 1987, pp. 331–339. Author-hosted scan; printed pp. 331–336 actually read. `https://users.fmf.uni-lj.si/klavzar/preprints/hereditary.pdf`.

[2] V. V. Lozin, C. Mayhill and V. Zamaraev, *A note on the speed of hereditary graph properties*, Electronic Journal of Combinatorics 18(1) (2011), P157, doi:10.37236/644. Printed pp. 1–2 define the labelled speed and hereditary property. `https://wrap.warwick.ac.uk/id/eprint/38522/1/WRAP_Lozin_v18i1p157.pdf`.